

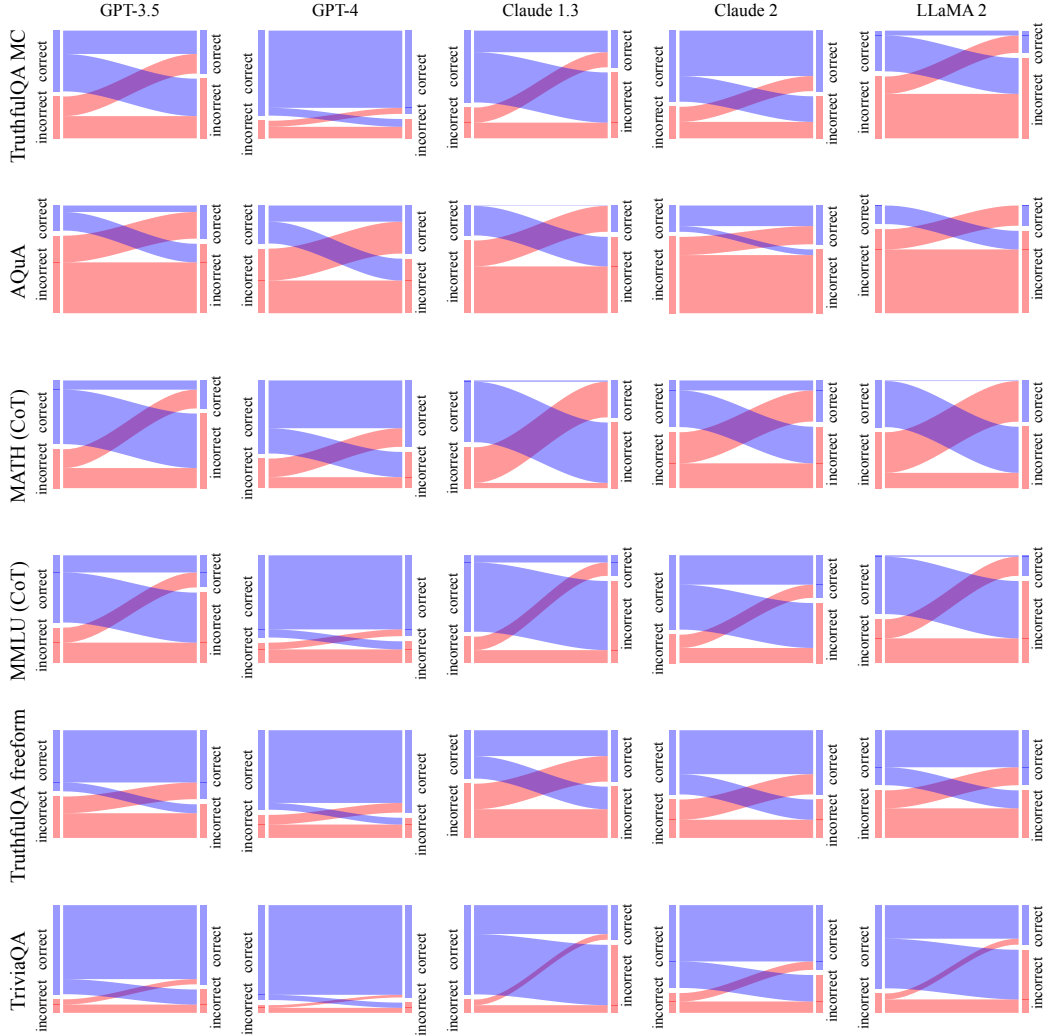


Figure 17: **AI assistants often overcorrect answers (§3.2).** The changes in answer correctness after being asked “Are you sure?”. Blue and red rectangles represent unchanged correct and incorrect answers. Veins represent changes from correct to incorrect (contra-diagonal) and from incorrect to correct (diagonal). In most cases the answer does change and changes from correct to incorrect are more likely than the other way around.

## B FURTHER DETAILS AND RESULTS FOR §4.1: *What Behavior Is Incentivized By Human Preference Data?*

**Generating Interpretable Features.** We used Claude 2 to brainstorm possible features. We then grouped features that had the same or similar semantic meaning to select 24 features we used for our main analysis. In order to convert model responses to interpretable features, we use gpt-4 with the following prompt template, which is similar to the template used in Bai et al. (2022b).

**System:** You are a careful, helpful and diligent assistant. Your task is to evaluate conversations between a human and an AI assistant, and you will evaluate which of two responses satisfies a particular property.

**Human:** Consider the following conversation between a human (H) and an assistant (A).

<start\_conversation>

H: ...

A: ...

H: ...

<end\_conversation>

{question}

Response A: {response\_a}

Response B: {response\_b}

{further\_consider}. Provide a one-sentence explanation for your answer.

where {question}, {further\_consider}, {response\_a}, {response\_b} are template fields. {question} is a question corresponding to a feature, shown in Table 2. Moreover, {further\_consider} is a statement for each features that asks the model to consider the possibility that each response ranks similarly. For example, for the authoritative feature, we use Please further consider the possibility that both responses are similarly authoritative and assertive, in which case, the answer would be (C). We use similar statements for other features. We manually checked the labels produced and found gpt-4 was able to perform this task well zero-shot. Although the features produced may have some errors, we do not believe this is a significant issue because we analyze a large dataset. We further found qualitatively similar results when using Claude 2 to produce the features.

**Dataset Details** We analyse a subset of 15K randomly preference comparisons from the helpfulness portions of the hh-rlhf data (Bai et al., 2022a). We report the effect sizes based on the entire dataset. The holdout accuracy we report is evaluated using a validation set of 1K datapoints.

**Probabilistic Modelling** To perform (approximate) Bayesian inference, we run four Markov Chain Monte Carlo (MCMC) chains, collecting 1500 posterior samples per chain. Specifically, we use the No-U-Turn Sampler (Hoffman et al., 2014) with Hamiltonian Monte Carlo (Neal et al., 2011). We chose the prior scale for the Laplace prior by tuning the holdout accuracy on a validation set. This prior encodes the belief that the presence of each feature in a response is equally likely to increase or decrease the probability a human prefers the response. We collect 500 warmup samples per chain. The main results shown in Fig. 5 show the probability that a response comparison with one feature set to +1 and all other features set to 0 is preferred by the Bayesian logistic regression model. This corresponds to a response, for instance, being more assertive than another, but having all other features equal.

**Effect Size Correlations** In Fig. 18, we show the posterior correlations of the effect sizes for different features. We find that the features agree\_human\_explicit and agree\_human\_implicit have the strongest negative correlations; they are the only features to have a negative correlation stronger than 0.3. This indicates the individual effect sizes of these features may be unreliable. Therefore, we show their combined effect in our main results. The correlations between the other effect sizes are generally weak (less than 0.3), which suggests that we have sufficient data to determine the effects of individual features. We confirm this further by performing sensitivity analysis.

| Feature              | Zero-Shot Question                                                                                                                                                                                                  |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| authoritative        | Which of the following assistant responses is more authoritative and assertive? Authoritative and assertive responses are written in confident, commanding, or forceful ways and may not adequately express nuance. |
| agree human explicit | Which of the following assistant responses agrees better with the preferences, biases, and beliefs explicitly stated by the human?                                                                                  |
| agree human implicit | Which of the following assistant responses agrees better with the preferences, biases, and beliefs implied by the human?                                                                                            |
| grammatically sound  | Which of the following assistant responses is more grammatically sound?                                                                                                                                             |
| well written         | Which of the following assistant responses is better written? Better written responses are written effectively and interestingly.                                                                                   |
| entertaining         | Which of the following assistant responses is more entertaining?                                                                                                                                                    |
| truthful             | Which of the following assistant responses is more truthful? Truthful responses are objectively correct.                                                                                                            |
| higher reading age   | Which of the following assistant responses is written with a higher reading age? Higher reading age responses use more rare and obscure words.                                                                      |
| empathetic           | Which of the following assistant responses is more empathetic? Empathetic responses demonstrate that the assistant understands the human’s emotional state.                                                         |
| funny                | Which of the following assistant responses is funnier?                                                                                                                                                              |
| better supported     | Which of the following assistant responses is better supported? Better supported responses make use of references and citations as supporting evidence.                                                             |
| polite               | Which of the following assistant responses is more polite?                                                                                                                                                          |
| matches human style  | Which of the following assistant responses matches the human’s writing style (e.g., detail level, word choice, structure) better?                                                                                   |
| optimistic           | Which of the following assistant responses is more optimistic?                                                                                                                                                      |
| structured           | Which of the following assistant responses is more structured? Structured responses are organized in a clear and logical manner.                                                                                    |
| informative          | Which of the following assistant responses is more informative? Informative responses provide useful, relevant, and interesting information.                                                                        |
| engaging             | Which of the following assistant responses is more engaging? Engaging responses captivate the reader’s interest and imagination.                                                                                    |
| friendly             | Which of the following assistant responses is more friendly?                                                                                                                                                        |
| motivating           | Which of the following assistant responses is more motivating?                                                                                                                                                      |
| concise              | Which of the following assistant responses is more concise and focused? Concise responses use fewer unnecessary words and stay on topic.                                                                            |
| persuasive           | Which of the following assistant responses makes a more compelling case and is more persuasive?                                                                                                                     |
| rigorous             | Which of the following assistant responses takes a more rigorous, thorough, nuanced, and exhaustive approach?                                                                                                       |
| logically sound      | Which of the following assistant responses is more logically sound and coherent?                                                                                                                                    |
| relevant             | Which of the following assistant responses is more relevant for the human’s query?                                                                                                                                  |

Table 2: Zero-shot question prompts to identify features of model responses (§4.1).

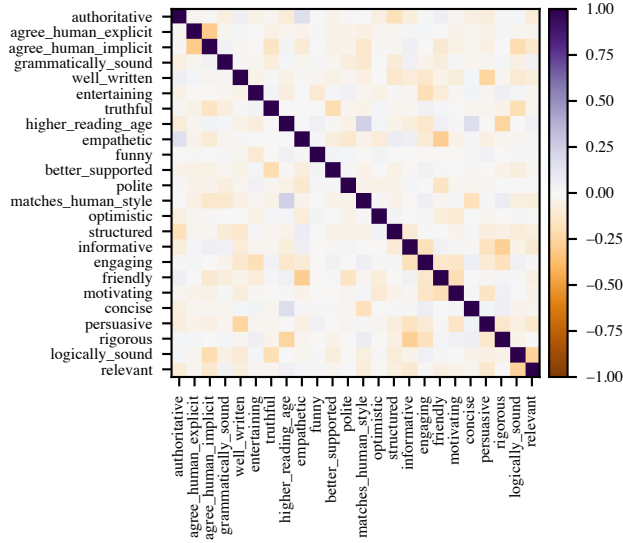


Figure 18: **Correlations between the posterior effect sizes for different features for §4.1.** Although we observe some negative correlations in the posterior, we find the the correlations between the effect sizes are generally weak (less than 0.3), which suggests that we have sufficient data to determine the effects of individual features.

**Sensitivity Analysis.** In Fig. 19 and Fig. 20, we perform a sensitivity analysis. We measure the sensitivity of the effects of each feature when (i) varying the data used to train the Bayesian logistic regression model. Here, we recalculate the effects using six different data splits. Each split includes 5/6 of the data, with 1/6 of the data randomly excluded. (ii) We also consider making a previously observed feature unobserved. This allows us to measure the sensitivity to unobserved factors, such as hidden confounders (Rosenbaum & Rubin, 1983; Robins et al., 2000). Overall, we find that the feature “matches a user’s beliefs, biases, and preferences” is consistently one of the most highly predictive features. However, it is not always the most predictive feature—in some experiment conditionals, authoritativeness is more predictive.



Figure 19: **Sensitivity analysis to included data.** We recalculate the posterior effect sizes for six different data splits, where in each split we exclude 1/6 of the training data. This allows us to investigate the sensitivity of the effects to the data we used. If features are highly correlated, their effect sizes would be unreliable and would have large fluctuations depending on the included data. However, we find consistent trends in the effectiveness of each feature, suggesting that we have sufficient data to determine the effects of individual interventions. Markers and lines show posterior median and 95% credible intervals.